

Probability and Statistical Inference

Technical reference

Companion to the note on lorenzoliuzzo.github.io. Statements, conditions and proofs of the results used there.

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1 Probability toolkit

1.1 Probability spaces and the axioms

A probability space is a triple $(\Omega, \mathcal{F}, P)$: a sample space Ω , a σ -algebra $\mathcal{F}$ of events (a family of subsets of Ω containing Ω and closed under complements and countable unions), and a function $P : \mathcal{F} \rightarrow [0, 1]$ satisfying the axioms below. Events A, B are *disjoint* if $A \cap B = \emptyset$; events $B_1, B_2, \dots$ form a *partition* if they are pairwise disjoint and $\bigcup_i B_i = \Omega$.

Definition (Kolmogorov axioms). (A1) $P(A) \geq 0$ for every event A . (A2) $P(\Omega) = 1$. (A3) For pairwise disjoint events $A_1, A_2, \dots$, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

Theorem (Consequences of the axioms). For events $A, B, A_1, A_2, \dots$: (i) $P(\emptyset) = 0$; (ii) finite additivity: $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ for disjoint A_i ; (iii) $P(A^c) = 1 - P(A)$; (iv) if $A \subseteq B$ then $P(A) \leq P(B)$; in particular $P(A) \leq 1$; (v) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; (vi) union bound: $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (i) Take $A_i = \emptyset$ for all i in (A3): $P(\emptyset) = \sum_i P(\emptyset)$, which forces $P(\emptyset) = 0$. (ii) Apply (A3) with $A_i = \emptyset$ for $i > n$ and use (i). (iii) $\Omega = A \cup A^c$ is a disjoint union, so $1 = P(A) + P(A^c)$. (iv) $B = A \cup (B \setminus A)$ is disjoint, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$; with $B = \Omega$ this gives $P(A) \leq 1$. (v) Write $A \cup B = A \cup (B \setminus A)$ and $B = (A \cap B) \cup (B \setminus A)$, both disjoint; then $P(B \setminus A) = P(B) - P(A \cap B)$, and substituting gives (v). (vi) Set $B_1 = A_1$ and $B_i = A_i \setminus \bigcup_{j < i} A_j$. The B_i are disjoint, $\bigcup_i B_i = \bigcup_i A_i$ and $B_i \subseteq A_i$, so by (A3) and (iv) $P(\bigcup_i A_i) = \sum_i P(B_i) \leq \sum_i P(A_i)$. $\square$

The general inclusion–exclusion formula is $P(\bigcup_{i=1}^n A_i) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots$. Two further facts follow from (A3) by taking limits: if A_n increases to A then $P(A_n) \rightarrow P(A)$, and likewise for decreasing sequences (continuity of P). When Ω is finite and outcomes are equally likely, $P(A) = |A| / |\Omega|$.

1.2 Conditional probability, independence, Bayes

Definition. For $P(B) > 0$, the conditional probability of A given B is $P(A | B) = P(A \cap B) / P(B)$. Then $P(\cdot | B)$ is itself a probability: it satisfies (A1)–(A3) on $\mathcal{F}$, so all the consequences above hold conditionally.

Rearranging gives the multiplication rule $P(A \cap B) = P(A | B) P(B)$ and, by induction, the chain rule

$$P(A_1 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) \dots P(A_n | A_1 \cap \dots \cap A_{n-1}). \quad (1)$$

Definition (Independence). A and B are independent if $P(A \cap B) = P(A) P(B)$ (equivalently $P(A | B) = P(A)$ when $P(B) > 0$). Events $A_1, \dots, A_n$ are *mutually independent* if $P(\bigcap_{i \in S} A_i) = \prod_{i \in S} P(A_i)$ for every subset S of indices; pairwise independence alone does not imply this. A and B are conditionally independent given C if $P(A \cap B | C) = P(A | C) P(B | C)$.

Remark. Disjoint events with positive probability are never independent: $P(A \cap B) = 0 \neq P(A) P(B)$.

Theorem (Law of total probability and Bayes' theorem). Let $B_1, B_2, \dots$ be a partition of Ω with $P(B_i) > 0$. For any event A ,

$$P(A) = \sum_i P(A | B_i) P(B_i), \quad (2)$$

and, if $P(A) > 0$,

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_i P(A | B_i) P(B_i)}. \quad (3)$$

Proof. The sets $A \cap B_i$ are disjoint with union A , so by (A3) and the multiplication rule $P(A) = \sum_i P(A \cap B_i) = \sum_i P(A | B_i) P(B_i)$. For Bayes, $P(B_j | A) = P(A \cap B_j) / P(A) = P(A | B_j) P(B_j) / P(A)$, and the denominator is Equation 2. $\square$

In Bayesian language $P(B_j)$ is the prior, $P(A | B_j)$ the likelihood, $P(A)$ the evidence (marginal likelihood) and $P(B_j | A)$ the posterior. The odds form is $P(B | A) / P(B^c | A) = [P(A | B) / P(A | B^c)] \cdot [P(B) / P(B^c)]$: posterior odds equal likelihood ratio times prior odds.

Example (Diagnostic test). Prevalence $P(D) = 0.01$, sensitivity $P(+ | D) = 0.95$, false-positive rate $P(+ | D^c) = 0.10$. Then

$$P(D | +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.10 \cdot 0.99} = \frac{0.0095}{0.1085} \approx 0.088. \quad (4)$$

The posterior is below 9% because the healthy majority produces more false positives than the sick minority produces true ones: the base rate matters.

1.3 Random variables and distributions

A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$. Its distribution function is $F(x) = P(X \leq x)$.

Lemma (Properties of the distribution function). F is non-decreasing and right-continuous, with $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$. For $a < b$, $P(a < X \leq b) = F(b) - F(a)$.

Proof. Monotonicity is (iv) applied to $\{X \leq x\} \subseteq \{X \leq y\}$. Right-continuity and the limits follow from continuity of P along monotone sequences of events. Finally $\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\}$ is disjoint. $\square$

X is discrete with mass function $p(x) = P(X = x)$, $\sum_x p(x) = 1$, or continuous with density $f = F'$, $\int f = 1$ and $P(a \leq X \leq b) = \int_a^b f$; in the continuous case $P(X = x) = 0$ for every x . The quantile function is $F^{-1}(p) = \inf\{x : F(x) \geq p\}$ and the median is $F^{-1}(1/2)$. For a pair (X, Y) with joint density $f_{X,Y}$, the marginal is $f_{X(x)} = \int f_{X,Y}(x, y) dy$, the conditional density is $f_{X|Y}(x | y) = f_{X,Y}(x, y)/f_{Y(y)}$, and X, Y are independent iff $f_{X,Y} = f_X f_Y$.

1.4 Expectation

Definition. $E[X] = \sum_x xp(x)$ (discrete) or $\int xf(x) dx$ (continuous), when the sum or integral converges absolutely. For a function g (law of the unconscious statistician), $E[g(X)] = \sum_x g(x)p(x)$ or $\int g(x)f(x) dx$.

Theorem (Properties of expectation). (i) Linearity: $E[aX + bY + c] = aE[X] + bE[Y] + c$ for any X, Y , dependent or not. (ii) Monotonicity: $X \leq Y$ implies $E[X] \leq E[Y]$. (iii) If X, Y are independent, $E[XY] = E[X]E[Y]$. (iv) $E[\mathbb{1}_A] = P(A)$. (v) Jensen: for convex g , $g(E[X]) \leq E[g(X)]$.

Proof. (i) and (ii) follow from linearity and monotonicity of sums and integrals, using the joint density for X and Y together. (iii) By independence $f_{X,Y} = f_X f_Y$, so $E[XY] = \int \int xy f_{X(x)} f_{Y(y)} dx dy = E[X]E[Y]$. (iv) $\mathbb{1}_A$ takes the value 1 with probability $P(A)$ and 0 otherwise. (v) A convex g lies above its tangent line at $m = E[X]$: $g(x) \geq g(m) + c(x - m)$ for some slope c ; take expectations. $\square$

1.5 Variance, covariance, correlation

Definition. $\text{Var}(X) = E[(X - \mu)^2]$ with $\mu = E[X]$; the standard deviation is $\sigma = \sqrt{\text{Var}(X)}$. $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$ and, when $\sigma_X, \sigma_Y > 0$, the correlation is $\rho_{XY} = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$. Moments: $E[X^k]$; central moments $E[(X - \mu)^k]$; skewness $E[(X - \mu)^3]/\sigma^3$ and kurtosis $E[(X - \mu)^4]/\sigma^4$ (equal to 3 for the normal).

Theorem (Properties of variance and covariance). (i) $\text{Var}(X) = E[X^2] - (E[X])^2 \geq 0$, with equality iff X is almost surely constant. (ii) $\text{Var}(aX + b) = a^2 \text{Var}(X)$. (iii) $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$, $\text{Cov}(X, X) = \text{Var}(X)$, and Cov is bilinear. (iv) $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$. (v) Independent X, Y have $\text{Cov}(X, Y) = 0$; the converse is false. (vi) $|\rho_{XY}| \leq 1$, with equality iff $Y = aX + b$ almost surely.

Proof. (i) Expand $(X - \mu)^2 = X^2 - 2\mu X + \mu^2$ and take expectations; nonnegativity holds because the integrand is ≥ 0 , and $E[(X - \mu)^2] = 0$ forces $X = \mu$ almost surely. (ii) $aX + b - E[aX + b] = a(X - \mu)$. (iii) Expand the product and use linearity; bilinearity is linearity in each argument. (iv) is (ii) and (iii) combined. (v) follows from (iii) and property (iii) of expectation. For the failure of the converse, let $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$: $\text{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$ since odd moments of a symmetric law vanish, yet Y is a

function of X . (vi) For every real t , $0 \leq \text{Var}(Y - tX) = \text{Var}(Y) - 2t \text{Cov}(X, Y) + t^2 \text{Var}(X)$. A quadratic in t that never goes negative has non-positive discriminant, $4\text{Cov}(X, Y)^2 - 4 \text{Var}(X) \text{Var}(Y) \leq 0$, which is $|\rho| \leq 1$. Equality means the discriminant vanishes, so $\text{Var}(Y - tX) = 0$ for some t and $Y - tX$ is constant. $\square$

Property (iv) generalizes to $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$, and so, for uncorrelated (hence independent) X_i with common variance σ^2 ,

$$\text{Var}\left(\sum_i X_i\right) = n\sigma^2, \quad \text{Var}(\bar{X}_n) = \sigma^2/n. \quad (5)$$

Theorem (Law of total expectation and total variance). If $E[X^2] < \infty$, then $E[X] = E[E[X | Y]]$ and

$$\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y]). \quad (6)$$

Proof. The first identity is the law of total probability for expectations: $E[X] = \int E[X | Y = y] f_{Y(y)} dy$. For the second, write $X - E[X] = (X - E[X | Y]) + (E[X | Y] - E[X])$ and square. The cross term has expectation zero: conditionally on Y the second factor is fixed and $E[X - E[X | Y] | Y] = 0$. The first square has expectation $E[\text{Var}(X | Y)]$ and the second $\text{Var}(E[X | Y])$. $\square$

1.6 Markov and Chebyshev

Lemma (Markov and Chebyshev). If $X \geq 0$ and $a > 0$ then $P(X \geq a) \leq E[X]/a$. For any X with mean μ and variance σ^2 and any $k > 0$, $P(|X - \mu| \geq k) \leq \sigma^2/k^2$.

Proof. $a \mathbb{1}\{X \geq a\} \leq X$ pointwise, take expectations. Chebyshev is Markov applied to $(X - \mu)^2$ and $a = k^2$. $\square$

1.7 Common distributions

Distribution	Mass or density	Mean	Variance
Bernoulli(p)	$p^x(1-p)^{1-x}$, $x \in \{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Poisson(λ)	$e^{-\lambda} \lambda^x / x!$	λ	λ
Uniform(a, b)	$1/(b-a)$ on $[a, b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$\lambda e^{-\lambda x}$, $x \geq 0$	$1/\lambda$	$1/\lambda^2$
Normal(μ, σ^2)	$(2\pi\sigma^2)^{-1/2} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2

1.8 Distributions used in inference

Definition (Gaussian family). $X \sim \mathcal{N}(\mu, \sigma^2)$ has density $(2\pi\sigma^2)^{-1/2} \exp(-(x-\mu)^2/(2\sigma^2))$. Let $Z_1, \dots, Z_k \sim \mathcal{N}(0, 1)$ be independent. Then

- $\sum_i Z_i^2 \sim \chi_k^2$ (chi-square, k degrees of freedom), with mean k and variance $2k$;
- if $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_k^2$ are independent, $Z/\sqrt{V/k} \sim t_k$ (Student);
- if $V_1 \sim \chi_a^2$ and $V_2 \sim \chi_b^2$ are independent, $(V_1/a)/(V_2/b) \sim F_{a,b}$.

Facts used below: a linear combination of jointly Gaussian variables is Gaussian; for jointly Gaussian variables, uncorrelated implies independent; $t_k \rightarrow \mathcal{N}(0, 1)$ as $k \rightarrow \infty$; $t_k^2 \sim F_{1,k}$.

1.9 Limit theorems

Write $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

Theorem (Weak law of large numbers). If X_i are i.i.d. with $E X_1 = \mu$ and $\text{Var}(X_1) = \sigma^2 < \infty$, then $\bar{X}_n \xrightarrow{P} \mu$, i.e. $P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

Proof. By Equation 5, $\text{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev gives $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$. $\square$

Theorem (Central limit theorem). If X_i are i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$, then

$$\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1). \quad (7)$$

The proof goes through characteristic functions and is not reproduced here. The Berry–Esseen theorem quantifies the rate: the Kolmogorov distance to Φ is at most $C E|X_1 - \mu|^3 / (\sigma^3 \sqrt{n})$. This is why the sample size needed depends on skewness.

Theorem (Slutsky and delta method). If $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{P} b$ (constant), then $A_n + B_n \xrightarrow{d} A + b$ and $A_n B_n \xrightarrow{d} bA$. If moreover $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \tau^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then

$$\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, g'(\theta)^2 \tau^2). \quad (8)$$

Slutsky is what licenses replacing an unknown σ by a consistent estimate in a pivot, which is how the Wald and large-sample t intervals are justified.

2 Sampling distributions of the normal model

Let $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d., with $\bar{X} = \bar{X}_n$ and $S^2 = (n-1)^{-1} \sum_i (X_i - \bar{X})^2$.

Theorem (Distribution of $\bar{X}$ and S^2). (i) $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$. (ii) $\bar{X}$ and S^2 are independent. (iii) $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$. (iv) Consequently

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}. \quad (9)$$

Proof. (i) is closure of the Gaussian family under linear maps. For (ii)–(iii), let $Z_i = (X_i - \mu)/\sigma$ and Q an orthogonal $n \times n$ matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$. Then $Y = QZ$ has $Y_1 = \sqrt{n}\bar{Z}$ and, since Q is orthogonal, $Y \sim \mathcal{N}(0, I_n)$: the Y_i are independent standard normals. Orthogonality preserves the norm: $\sum_i Z_i^2 = \sum_i Y_i^2$, and $\sum_i Z_i^2 = \sum_i (Z_i - \bar{Z})^2 + n\bar{Z}^2$, hence

$$\sum_i (Z_i - \bar{Z})^2 = \sum_{i=2}^n Y_i^2 \sim \chi_{n-1}^2, \quad (10)$$

a function of $(Y_2, \dots, Y_n)$ only, and so independent of Y_1 and thus of $\bar{Z}$. Since $\sum_i (Z_i - \bar{Z})^2 = (n-1)S^2/\sigma^2$ and $\bar{X} = \mu + \sigma\bar{Z}$, (ii) and (iii) follow. For (iv), write $T = [(\bar{X} - \mu)/(\sigma/\sqrt{n})] / \sqrt{((n-1)S^2/\sigma^2)/(n-1)}$, a standard normal over the root of an independent $\chi_{n-1}^2/(n-1)$. $\square$

Theorem (Unbiasedness of S^2). For any i.i.d. sample with finite variance σ^2 , $E[S^2] = \sigma^2$.

Proof. $\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Taking expectations, $E[\sum_i (X_i - \bar{X})^2] = n\sigma^2 - n(\sigma^2/n) = (n-1)\sigma^2$. $\square$

3 Point estimation

Let $X_1, \dots, X_n \sim f(\cdot; \theta)$, $\theta \in \Theta \subset \mathbb{R}$, and $\hat{\theta} = T(X_1, \dots, X_n)$.

Definition. $\text{bias}(\hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}] - \theta$. The estimator is unbiased if this vanishes for all θ . It is consistent if $\hat{\theta}_n \xrightarrow{P} \theta$. Its mean squared error is $\text{MSE}(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2]$.

Theorem (Bias–variance decomposition).

$$\text{MSE}(\hat{\theta}) = \text{Var}_\theta(\hat{\theta}) + \text{bias}(\hat{\theta})^2. \quad (11)$$

Proof. Add and subtract $\mathbb{E} \hat{\theta}$: $(\hat{\theta} - \theta)^2 = (\hat{\theta} - \mathbb{E} \hat{\theta})^2 + 2(\hat{\theta} - \mathbb{E} \hat{\theta})(\mathbb{E} \hat{\theta} - \theta) + (\mathbb{E} \hat{\theta} - \theta)^2$. The cross term has zero expectation because $\mathbb{E} \hat{\theta} - \theta$ is a constant and $\mathbb{E}[\hat{\theta} - \mathbb{E} \hat{\theta}] = 0$. $\square$

Example (A biased estimator with smaller MSE). For $X_i \sim \mathcal{N}(\mu, \sigma^2)$, the estimator $\hat{\sigma}_c^2 = c \sum_i (X_i - \bar{X})^2$ has, using $\sum_i (X_i - \bar{X})^2 \sim \sigma^2 \chi_{n-1}^2$ (mean $(n-1)\sigma^2$, variance $2(n-1)\sigma^4$),

$$\text{MSE} = \sigma^4[2c^2(n-1) + ((n-1)c - 1)^2]. \quad (12)$$

This is minimized at $c = 1/(n+1)$, not at the unbiased choice $c = 1/(n-1)$. The unbiased estimator is therefore not MSE-optimal, a first instance of the shrinkage idea behind Ridge and Lasso.

3.1 Fisher information and the Cramér–Rao bound

Assume the regularity conditions: the support of $f(\cdot; \theta)$ does not depend on θ , $\log f$ is twice differentiable in θ , and differentiation under the integral sign is allowed.

Definition (Score and Fisher information). The score of one observation is $s(x; \theta) = \partial_\theta \log f(x; \theta)$, and

$$I(\theta) = \mathbb{E}_\theta[s(X; \theta)^2] = \text{Var}_\theta(s(X; \theta)). \quad (13)$$

Lemma. Under the regularity conditions, $\mathbb{E}_\theta[s(X; \theta)] = 0$ and $I(\theta) = -\mathbb{E}_\theta[\partial_\theta^2 \log f(X; \theta)]$. For n independent observations the information is $nI(\theta)$.

Proof. $\mathbb{E}[s] = \int (\partial_\theta f / f) f dx = \partial_\theta \int f dx = \partial_\theta 1 = 0$. Differentiating $\int s f dx = 0$ once more gives $\int (\partial_\theta s) f dx + \int s^2 f dx = 0$, and $\partial_\theta s = \partial_\theta^2 \log f$. Additivity follows because the log-likelihood of independent data is the sum of the individual ones, so the scores add and, being independent with mean zero, their variances add. $\square$

Theorem (Cramér–Rao lower bound). If $\hat{\theta}$ is unbiased with finite variance, then $\text{Var}_\theta(\hat{\theta}) \geq 1/(nI(\theta))$.

Proof. Let $U = \sum_i s(X_i; \theta)$ be the total score, so $\mathbb{E} U = 0$ and $\text{Var}(U) = nI(\theta)$. Unbiasedness reads $\int \hat{\theta}(x) f_{n(x; \theta)} dx = \theta$ (f_n the joint density). Differentiating in θ gives $\int \hat{\theta}(x) U(x) f_n dx = 1$, i.e. $\text{Cov}(\hat{\theta}, U) = \mathbb{E}[\hat{\theta} U] = 1$ since $\mathbb{E} U = 0$. Cauchy–Schwarz: $1 = \text{Cov}(\hat{\theta}, U)^2 \leq \text{Var}(\hat{\theta}) \text{Var}(U) = \text{Var}(\hat{\theta}) nI(\theta)$. $\square$

Remark. For a biased estimator with $b(\theta) = \text{bias}(\hat{\theta})$ the same argument gives $\text{Var}(\hat{\theta}) \geq (1 + b'(\theta))^2 / (nI(\theta))$. Equality in Cauchy–Schwarz holds iff $\hat{\theta} - \theta$ is proportional to the score, which happens exactly in exponential families for the mean-value parameter.

3.2 Sufficiency (brief)

A statistic $T(X)$ is sufficient for θ if the conditional law of the sample given T does not depend on θ . By the Fisher–Neyman factorization theorem, T is sufficient iff $f_{n(x; \theta)} = g(T(x); \theta)h(x)$. For normal data with known σ , $\sum_i X_i$ is sufficient for μ ; for the Bernoulli model, $\sum_i X_i$ is sufficient for p . The Rao–Blackwell theorem states that conditioning an estimator on a sufficient statistic never increases its MSE.

4 Maximum likelihood

Definition. The likelihood is $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$, the log-likelihood $\ell(\theta) = \log L(\theta)$ and the maximum likelihood estimator $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta)$.

Example (Bernoulli). $\ell(p) = (\sum x_i) \log p + (n - \sum x_i) \log(1 - p)$, $\ell'(p) = (\sum x_i)/p - (n - \sum x_i)/(1 - p)$. Setting it to zero gives $\hat{p} = \bar{x}$; $\ell'' < 0$ so it is a maximum. The information is $I(p) = 1/(p(1 - p))$ and $\text{Var}(\hat{p}) = p(1 - p)/n = 1/(nI(p))$: the MLE attains the Cramér–Rao bound exactly.

Example (Normal). $\ell(\mu, \sigma^2) = -(n/2) \log(2\pi\sigma^2) - (2\sigma^2)^{-1} \sum_i (x_i - \mu)^2$. The first-order conditions give $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = n^{-1} \sum_i (x_i - \bar{x})^2$. By the theorem on S^2 , $E \hat{\sigma}^2 = ((n - 1)/n)\sigma^2$: the MLE of the variance is biased, but consistent, and $\hat{\sigma}^2 = cS^2$ with $c = (n - 1)/n \rightarrow 1$.

4.1 Asymptotic properties

Theorem (Asymptotics of the MLE). Under the regularity conditions above, with θ_0 the true parameter and $I(\theta_0) > 0$: (i) $\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta_0$; (ii) $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$; (iii) invariance: the MLE of $g(\theta)$ is $g(\hat{\theta}_{\text{MLE}})$, and by the delta method it is asymptotically $\mathcal{N}(g(\theta_0), g'(\theta_0)^2/(nI(\theta_0)))$.

Proof. Sketch of (ii). Expand the score at the MLE around θ_0 : $0 = \ell'(\hat{\theta}) = \ell'(\theta_0) + (\hat{\theta} - \theta_0)\ell''(\tilde{\theta})$ for some $\tilde{\theta}$ between them. Hence

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{n^{-1/2}\ell'(\theta_0)}{-n^{-1}\ell''(\tilde{\theta})}. \quad (14)$$

The numerator is $n^{-1/2} \sum_i s(X_i; \theta_0)$, a normalized sum of i.i.d. mean-zero variables with variance $I(\theta_0)$, so by the CLT it tends to $\mathcal{N}(0, I(\theta_0))$. By the law of large numbers and consistency, the denominator tends to $-E[\partial_{\theta}^2 \log f] = I(\theta_0)$. Slutsky gives the limit $\mathcal{N}(0, I(\theta_0)^{-1})$. $\square$

The practical consequence is the standard error $\hat{\text{se}} = 1/\sqrt{nI(\hat{\theta})}$ (or its observed-information version, $1/\sqrt{-\ell''(\hat{\theta})}$), valid for large n .

4.2 Method of moments

Equate the first d population moments $m_{j(\theta)} = E_{\theta} X^j$ to the sample moments $n^{-1} \sum_i X_i^j$ and solve for $\theta \in \mathbb{R}^d$. The estimator is consistent when $m = (m_1, \dots, m_d)$ is invertible with continuous inverse, but generally has larger asymptotic variance than the MLE.

5 Interval estimation

Definition. A $(1 - \alpha)$ confidence interval is a random interval $[L(X), U(X)]$ with $P_{\theta}(L(X) \leq \theta \leq U(X)) \geq 1 - \alpha$ for all $\theta \in \Theta$. The left side is the coverage; the inequality is replaced by an equality for exact intervals.

The probability refers to the random endpoints under repeated sampling. For the realized interval, θ is fixed and the event either holds or not.

Definition (Pivot). $Q(X; \theta)$ is a pivot if its distribution does not depend on θ .

If $P(a \leq Q(X; \theta) \leq b) = 1 - \alpha$ and Q is monotone in θ , solving $a \leq Q \leq b$ for θ gives an exact interval.

Theorem (Intervals for the normal model). Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d.. Exact $(1 - \alpha)$ intervals:

- mean, σ unknown: $\bar{X} \pm t_{n-1, 1-\alpha/2} S / \sqrt{n}$;
- mean, σ known: $\bar{X} \pm z_{1-\alpha/2} \sigma / \sqrt{n}$;
 $\left[(n-1)S^2 / \chi_{n-1, 1-\alpha/2}^2, \right.$
- variance: $\left. (n-1)S^2 / \chi_{n-1, \alpha/2}^2 \right]$.

Proof. $P(-t \leq T \leq t) = 1 - \alpha$ with $t = t_{n-1, 1-\alpha/2}$ and T the pivot of the sampling-distribution theorem; rearranging $-t \leq (\bar{X} - \mu) / (S / \sqrt{n}) \leq t$ gives the first interval. The second is identical with T replaced by $\sqrt{n}(\bar{X} - \mu) / \sigma \sim \mathcal{N}(0, 1)$. For the third, $P(\chi_{n-1, \alpha/2}^2 \leq (n-1)S^2 / \sigma^2 \leq \chi_{n-1, 1-\alpha/2}^2) = 1 - \alpha$ and inverting in σ^2 reverses the inequalities. $\square$

Theorem (Wald interval). If $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta))$ with v continuous, then $\hat{\theta} \pm z_{1-\alpha/2} \sqrt{v(\hat{\theta})/n}$ has asymptotic coverage $1 - \alpha$. For the MLE, $v = I^{-1}$. For a proportion, $\hat{p} \pm z_{1-\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$.

Proof. $\sqrt{n}(\hat{\theta} - \theta) / \sqrt{v(\hat{\theta})} = [\sqrt{n}(\hat{\theta} - \theta) / \sqrt{v(\theta)}] \cdot [\sqrt{v(\theta)/v(\hat{\theta})}]$; the first factor tends in distribution to $\mathcal{N}(0, 1)$ and the second in probability to 1 by consistency and continuity. Apply Slutsky. $\square$

Remark. The Wald interval for a proportion has coverage that oscillates and can be far below $1 - \alpha$ when n is small or p is near 0 or 1. The Wilson interval, obtained by inverting the score test instead of replacing p by $\hat{p}$ in the variance, is the standard fix.

6 Hypothesis testing

6.1 Framework

A test of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ is a rule $\varphi(X) \in \{0, 1\}$ (1 means reject), typically $\varphi = \mathbb{1}\{T(X) \in R\}$ for a statistic T and rejection region R .

Definition. The size of the test is $\sup_{\theta \in \Theta_0} P_\theta(\varphi = 1)$; it has level α if the size is at most α . The power function is $\beta(\theta) = P_\theta(\varphi = 1)$, so the Type II error at $\theta \in \Theta_1$ is $1 - \beta(\theta)$.

Definition. The p -value of observed t_{obs} for a statistic T that rejects for large values is $p = \sup_{\theta \in \Theta_0} P_\theta(T \geq t_{\text{obs}})$.

Lemma (Uniformity of the p -value). If H_0 is simple, T has a continuous distribution function G under H_0 , and rejection is for large T , then $p(T) = 1 - G(T) \sim \text{Uniform}(0, 1)$ under H_0 . Hence rejecting when $p \leq \alpha$ has size exactly α .

Proof. $G(T) \sim \text{Uniform}(0, 1)$ by the probability integral transform: $P(G(T) \leq u) = P(T \leq G^{-1}(u)) = u$. Then $1 - G(T)$ is also uniform. $\square$

6.2 The t -tests

Theorem (One-sample and Welch t -tests). (i) For $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d., under $H_0 : \mu = \mu_0$ the statistic $T = (\bar{X} - \mu_0) / (S / \sqrt{n})$ has the t_{n-1} distribution, so rejecting for $|T| > t_{n-1, 1-\alpha/2}$ has size α . (ii) For two independent normal samples with means μ_1, μ_2 , and $H_0 : \mu_1 = \mu_2$,

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \quad (15)$$

is approximately t_ν with

$$\nu = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{(S_1^2/n_1)^2/(n_1 - 1) + (S_2^2/n_2)^2/(n_2 - 1)}. \quad (16)$$

Part (i) is the pivot theorem with $\mu = \mu_0$. In (ii) the denominator is a weighted sum of independent scaled χ^2 variables, which is not exactly a scaled χ^2 ; the Welch–Satterthwaite ν matches the first two moments of a scaled χ_ν^2 to it. With equal variances the pooled test is exactly $t_{n_1+n_2-2}$, but Welch is robust to unequal variances at little cost.

Theorem (Duality of tests and intervals). If, for each θ_0 , $A(\theta_0)$ is the acceptance region of a level- α test of $\theta = \theta_0$, then $C(x) = \{\theta_0 : x \in A(\theta_0)\}$ is a $(1 - \alpha)$ confidence set. Conversely, from a confidence set one obtains a test by rejecting $\theta = \theta_0$ iff $\theta_0 \notin C(x)$.

Proof. $P_{\theta_0}(\theta_0 \in C(X)) = P_{\theta_0}(X \in A(\theta_0)) \geq 1 - \alpha$, for every θ_0 . The converse is the same identity read backwards. $\square$

6.3 Power and sample size

Theorem (Power of the one-sided z -test). Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with σ known, and reject $H_0 : \mu = \mu_0$ in favour of $H_1 : \mu > \mu_0$ when $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma > z_{1-\alpha}$. Then

$$\beta(\mu_1) = 1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu_1 - \mu_0)/\sigma), \quad (17)$$

and $\beta(\mu_1) \geq 1 - \beta_0$ holds iff $n \geq ((z_{1-\alpha} + z_{1-\beta_0})\sigma/(\mu_1 - \mu_0))^2$.

Proof. Under $\mu = \mu_1$, $\sqrt{n}(\bar{X} - \mu_1)/\sigma \sim \mathcal{N}(0, 1)$, so $Z = Z_0 + \sqrt{n}(\mu_1 - \mu_0)/\sigma$ with $Z_0 \sim \mathcal{N}(0, 1)$. Hence $P(Z > z_{1-\alpha}) = 1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu_1 - \mu_0)/\sigma)$. Requiring this to be at least $1 - \beta_0$ means $z_{1-\alpha} - \sqrt{n}\delta/\sigma \leq -z_{1-\beta_0}$ (with $\delta = \mu_1 - \mu_0$), and solving for n gives the bound. $\square$

The required n scales like σ^2/δ^2 : halving the detectable effect quadruples the sample.

6.4 Likelihood ratio tests

Theorem (Neyman–Pearson lemma). For simple $H_0 : \theta = \theta_0$ against simple $H_1 : \theta = \theta_1$, let φ^* reject when $L(\theta_1)/L(\theta_0) > k$, with k chosen so that φ^* has size α . Then for any other test φ of size at most α , $\beta_\varphi(\theta_1) \leq \beta_{\varphi^*}(\theta_1)$.

Proof. For every x , $(\varphi^*(x) - \varphi(x))(f(x; \theta_1) - kf(x; \theta_0)) \geq 0$: where $\varphi^* = 1$ the second factor is positive, where $\varphi^* = 0$ it is non-positive and the first factor is $-\varphi(x) \leq 0$ or zero. Integrating over x gives $\beta_{\varphi^*}(\theta_1) - \beta_\varphi(\theta_1) \geq k(\alpha_{\varphi^*} - \alpha_\varphi) \geq 0$, since both sizes are at most α and φ^* has size exactly α . $\square$

Theorem (Wilks' theorem). Let $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L$, where Θ has d free parameters and Θ_0 has $d - r$. Under the regularity conditions and H_0 ,

$$-2 \log \Lambda \xrightarrow{d} \chi_r^2. \quad (18)$$

The proof expands ℓ quadratically around the MLE and uses asymptotic normality of the MLE, so that $-2 \log \Lambda$ becomes a quadratic form in r asymptotically independent standard normals. The Wald and score tests are asymptotically equivalent to it. In the linear model with Gaussian errors, the likelihood ratio is a monotone function of an F statistic, which is why the F -test of regression has an exact distribution.

6.5 Multiple testing

With m tests and all nulls true, each at level α , the expected number of false rejections is $m\alpha$.

Theorem (Bonferroni and Benjamini–Hochberg). (i) Rejecting each hypothesis with $p_i \leq \alpha/m$ controls the family-wise error rate, $P(\text{at least one false rejection}) \leq \alpha$, with no assumption on the dependence of the p -values. (ii) Order the p -values $p_{(1)} \leq \dots \leq p_{(m)}$ and reject the k smallest, where $k = \max\{i : p_{(i)} \leq i\alpha/m\}$. If the p -values are independent, the false discovery rate $E[V/\max(R, 1)]$ (with V false rejections out of R) is at most $\alpha m_0/m \leq \alpha$, where m_0 is the number of true nulls.

Proof. (i) By the union bound, $P\left(\bigcup_{i \in H_0} \{p_i \leq \alpha/m\}\right) \leq m_0\alpha/m \leq \alpha$. (ii) is the Benjamini–Hochberg (1995) theorem; the proof conditions on the other p -values and is omitted. $\square$

7 Summary of the main formulas

Quantity	Result
Standard error of the mean	$\sigma/\sqrt{n}$
S^2 unbiased	$E S^2 = \sigma^2$, divisor $n - 1$
Normal sampling laws	$(n - 1)S^2/\sigma^2 \sim \chi_{n-1}^2$, $T \sim t_{n-1}$, $\bar{X} \perp S^2$
MSE	$\text{Var} + \text{bias}^2$
Cramér–Rao	$\text{Var}(\hat{\theta}) \geq 1/(nI(\theta))$
MLE	$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$
CI for the mean	$\bar{X} \pm t_{n-1, 1-\alpha/2} S/\sqrt{n}$
CI for the variance	$[(n - 1)S^2/\chi_{n-1, 1-\alpha/2}^2, (n - 1)S^2/\chi_{n-1, \alpha/2}^2]$
Wald interval	$\hat{\theta} \pm z_{1-\alpha/2} \hat{\text{se}}$
Sample size, one-sided z -test	$n = ((z_{1-\alpha} + z_{1-\beta})\sigma/\delta)^2$
Likelihood ratio	$-2 \log \Lambda \xrightarrow{d} \chi_r^2$
Bonferroni / BH	$p_i \leq \alpha/m$ / largest k with $p_{(k)} \leq k\alpha/m$

8 References

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